1) a)

$x_0 = 2, y_0 = 1$

$y_1 = 1 + 0.5\text{f}(2)$

$= 1 + 0.5 \times 0.0183156889 \ldots = 1.009157819 \ldots$

$x_1 = 2 + 0.5 = 2.5$

$y_2 = y_1 + 0.5\text{f}(x_1) = 1.009\,157\,819 \ldots + 0.5 \times \text{f}(2.5)$

$= 1.009157819 \ldots + 0.5 \times 0.001930454136 \ldots$

$= 1.01012304 \ldots$

$x_2 = 2.5 + 0.5 = 3$

$\therefore y = 1.010$ (3 DP)

1) b)

$x_0 = 2, y_0 = 1$

$y_1 = 1 + 0.25\text{f}(2)$

$= 1 + 0.25 \times 0.018315688 \ldots = 1.00457890972 \ldots$

$x_1 = 2 + 0.25 = 2.25$

$y_2 = y_1 + 0.25\text{f}(x_1) = 1.00457890972 \ldots + 0.25 \times \text{f}(2.25)$

$= 1.00457890972 \ldots + 0.25 \times 0.00632971542749 \ldots$

$= 1.006161613 \ldots$

$x_2 = 2.25 + 0.25 = 2.5$

$y_3 = y_2 + 0.25\text{f}(x_2) = 1.006161613 \ldots + 0.25 \times \text{f}(2.5)$

$= 1.006161613 \ldots + 0.25 \times 0.001930454 \ldots$

$= 1.00664395211 \ldots$

$x_3 = 2.5 + 0.25 = 2.75$

$y_4 = y_3 + 0.25\text{f}(x_3) = 1.00664395211 \ldots + 0.25 \times \text{f}(2.75)$

$= 1.00664395211 \ldots + 0.25 \times 0.00051957 \ldots$

$= 1.00677384578 \ldots$

$x_4 = 2.75 + 0.25 = 3$

$\therefore y = 1.007$ (3 DP)

2) a)

$x_0 = 0, y_0 = 1$

$y_1 = y_0 + hf(x_0, y_0) = 1 + 0.2 \times f(0,1)$

$= 1 + 0.2 \times -1 = 0.8$

$x_1 = 0 + 0.2 = 0.2$

$y_2 = y_1 + hf(x_1) = 0.8 + 0.2 \times f(0.2, 0.8)$

$= 0.8 + 0.2 \times -0.76$

$= 0.648$

$x_2 = 0.2 + 0.2 = 0.4$

$\therefore y = 0.648$ (3 DP)

2) b)

$x_0 = 0, y_0 = 1$

$y_1 = y_0 + hf(x_0, y_0) = 1 + hf(0,1)$

$= 1 + 0.1 \times -1 = 0.9$

$x_1 = 0 + 0.1 = 0.1$

$y_2 = y_1 + hf(x_1, y_1) = 0.9 + 0.1 \times f(0.1, 0.9)$

$= 0.9 + 0.1 \times -0.89 = 0.811$

$x_2 = 0.1 + 0.1 = 0.2$

$y_3 = y_2 + hf(x_2, y_2) = 0.811 + 0.1 \times f(0.2, 0.811)$

$= 0.811 + 0.1 \times -0.771 = 0.7339$

$x_3 = 0.2 + 0.1 = 0.3$

$y_4 = y_3 + hf(x_3, y_3) = 0.7339 + 0.1 \times f(0.3, 0.7339)$

$= 0.7339 + 0.1 \times -0.6439 = 0.66951$

$x_4 = 0.3 + 0.1 = 0.4$

$\therefore y = 0.670$ (3 DP)

3) a)

$x_0 = 0, y_0 = 2$

$y_1 = y_0 + hf(x_0, y_0) = 2 + 0.2 \times f(0, 2)$

$= 2 + 0.2 \times -2 \ = 1.6$

$x_1 = 0 + 0.2 = 0.2$

$y_2 = y_1 + hf(x_1, y_1) = 1.6 + 0.2 \times f(0.2, 1.6)$

$= 1.6 + 0.2 \times -1.4013306692 \ldots = 1.3197338662 \ldots$

$x_2 = 0.2 + 0.2 = 0.4$

$\therefore y = 1.320$ (3 DP)

3) b)

$x_0 = 0, y_0 = 2$

$y_1 = y_0 + hf(x_0, y_0) = 2 + 0.1 \times f(0, 2)$

$= 2 + 0.1 \times -2 \ = 1.8$

$x_1 = 0 + 0.1 = 0.1$

$y_2 = y_1 + hf(x_1, y_1) = 1.8 \ + 0.1 \times f(0.1, 1.8)$

$= 1.8 \ + 0.1 \times -1.7001665834 \ldots = 1.6299833417 \ldots$

$x_2 = 0.1 + 0.1 = 0.2$

$y_3 = y_2 + hf(x_2, y_2) = 1.6299833417 \ldots + 0.1 \times f(0.2, 1.6299833417 \ldots)$

$= 1.6299833417 \ldots + 0.1 \times -1.4313140109 \ldots \ = 1.486851941 \ldots$

$x_3 = 0.2 + 0.1 = 0.3$

$y_4 = y_3 + hf(x_3, y_3) = 1.486851941 \ldots + 0.1 \times f(0.3, 1.486851941 \ldots)$

$= 1.486851941 \ldots + 0.1 \times -1.1913317339 \ = 1.3677187672 \ldots$

$x_4 = 0.3 + 0.1 = 0.4$

$\therefore y = 1.368$ (3 DP)

4)

$x_0 = 2, y_0 = -1$ so want $x = 3$ in two steps: $h = 0.5$

$y_1 = y_0 + hf(x_0, y_0) = -1 + 0.5 \times f(2, -1)$

$= -1 + 0.5 \times -3 = -2.5$

$x_1 = 2 + 0.5 = 2.5$

$y_2 = y_1 + hf(x_1, y_1) = -2.5 + 0.5 \times f(2.5, -2.5)$

$= -2.5 + 0.5 \times 1 = -2$

$x_2 = 2.5 + 0.5 = 3$

$\therefore y = -2$

5) a)

$\frac{dy}{dx} = \frac{e^{\sqrt{x}}}{2x}$

$x_0 = 4,\ y_0 = 1$

$y_1 = y_0 + hf(x_0) = 1 + 0.5 \times f(x_0)$

$= 1 + 0.5 \times 0.9236320124 \ldots = 1.46186006 \ldots$

$x_1 = 4 + 0.5 = 4.5$

$y_2 = y_1 + hf(x_1) = 1.46186006 \ldots + 0.5 \times f(x_1)$

$= 1.46186006 \ldots + 0.5 \times 0.9269049685 \ldots = 1.92526849 \ldots$

$x_2 = 4.5 + 0.5 = 5$

$\therefore y = 1.925$ (3 DP)

5) b)

$\frac{dy}{dx} = \frac{\ln(x^2+y)}{y}$

$x_0 = 0,\ y_0 = 2$

$y_1 = y_0 + hf(x_0, y_0) = 2 + 0.5 \times f(0, 2)$

$= 2 + 0.5 \times 0.3465735903 \ldots = 2.1732867951 \ldots$

$x_1 = 0 + 0.5 = 0.5$

$y_2 = y_1 + hf(x_1, y_1) = 2.1732867951 \ldots + 0.5 \times f(0.5, 2.1732867951 \ldots)$

$= 2.1732867951 \ldots + 0.5 \times 0.4072747327 \ldots = 2.3769241615 \ldots$

$x_2 = 0.5 + 0.5 = 1$

$y_3 = y_2 + hf(x_2, y_2) = 2.3769241615 \ldots + 0.5 \times f(1, 2.3769241615 \ldots)$

$= 2.3769241615 \ldots + 0.5 \times 0.5119916334 \ldots = 2.6329199781 \ldots.$

$x_3 = 1 + 0.5 = 1.5$

$\therefore y = 2.633$ (3 DP)

6) a)

$x_0 = 0, y_0 = 1$

$y_1 = y_0 + hf(x_0) = 1 + 0.2 \times f(0)$

$= 1 + 0.2 \times 2 = 1.4$

$x_1 = 0 + 0.2 = 0.2$

$y_2 = y_1 + hf(x_1) = 1.4 + 0.2 \times f(0.2)$

$= 1.4 + 0.2 \times 2e^{0.4} \ldots = 1.996729879 \ldots$

$x_2 = 0.2 + 0.2 = 0.4$

$y_3 = y_2 + hf(x_2) = 1.996729879 \ldots + 0.2 \times f(0.4)$

$= 1.996729879 \ldots + 0.2 \times 2e^{0.8} \ldots = 2.88694625 \ldots.$

$x_3 = 0.4 + 0.2 = 0.6$

$\therefore y = 2.887$ (3 DP)

6) b)

$\int 1 \, dy = \int 2e^{2x} \, dx$

$y = e^{2x} + c$

Given $y(0) = 1$ then $1 = e^0 + c \therefore c = 0$

$y = e^{2x}$

When $x = 0.6$, $y = e^{1.2}$

6) c)

$\frac{2.887 - e^{1.2}}{e^{1.2}} = -0.130$

6) d)

The relative error is negative, which means the approximate value is less than the exact value. Therefore, the Euler approximation underestimates the true value of $y(0.6)$

7) a)

$x_0 = 1, y_0 = 4$

$y_1 = y_0 + h\mathrm{f}(x_0, y_0) = 4 + 0.4 \times \mathrm{f}(1, 4)$

$= 4 + 0.4 \times -0.9609308626 \ldots = 3.6156276550 \ldots$

$x_1 = 1 + 0.4 = 1.4$

$y_2 = y_1 + h\mathrm{f}(x_1, y_1) = 3.6156276550 \ldots + 0.4 \times \mathrm{f}(1.4, 3.6156276550 \ldots)$

$= 3.6156276550 \ldots + 0.4 \times -1.2692009154 \ldots = 3.1079472888 \ldots$

$x_2 = 1.4 + 0.4 = 1.8$

$y_3 = y_2 + h\mathrm{f}(x_2, y_2) = 3.1079472888 \ldots + 0.4 \times \mathrm{f}(1.8, 3.1079472888 \ldots)$

$= 3.1079472888 \ldots + 0.4 \times -1.1983021443 \ldots = 2.6286264311 \ldots.$

$x_3 = 1.8 + 0.4 = 2.2$

$\therefore y = 2.629$ (3 DP)

7) b)

Not very reliable as Euler's step by step method assumes the gradient is constant over each step, which is not true here since $\cos y$ changes rapidly.

7) c)

Increase the number of steps (use a smaller value for h)

8)

$t_0 = 0, C_0 = 0$

$C_1 = C_0 + h\mathrm{f}(t_0, C_0) = 0 + 0.5 \times \mathrm{f}(0, 0)$

$= 0 + 0.5 \times 5 = 2.5$

$t_1 = 0 + 0.5 = 0.5$

$C_2 = C_1 + h\mathrm{f}(t_1, C_1) = 2.5 + 0.5 \times \mathrm{f}(0.5, 2.5)$

$= 2.5 + 0.5 \times 4.55 = 4.775$

$t_2 = 0.5 + 0.5 = 1$

$\therefore C \approx 4.775$ mg

9)

$x_0 = 0, y_0 = 12000$
$y_1 = y_0 + hf(x_0, y_0) = 12000 + 1 \times f(0, 12000)$
$= 12000 + 1 \times 60 \ = 12060$
$x_1 = 0 + 1 = 1$
$y_2 = y_1 + hf(x_1, y_1) = 12060 + 1 \times f(1, 12060)$
$= 12060 + 1 \times 160.3 = 12220.3$
$x_2 = 1 + 1 = 2$
$y_3 = y_2 + hf(x_2, y_2) = 12220.3 + 1 \times f(2, 12220.3)$
$= 12220.3 + 1 \times 234.3065808 \ldots = 12454.6065808 \ldots$
$x_3 = 2 + 1 = 3$
$\therefore y = 12455$ people

10)

$t_0 = 5, T_0 = 50$
$T_1 = T_0 + hf(t_0, T_0) = 50 + 1 \times f(5, 50)$
$= 50 + 5 \times -1.5 \ = 48.5$
$t_1 = 5 + 1 = 6$
$T_2 = T_1 + hf(t_1, T_1) = \ 48.5 + 1 \times f(6, 48.5)$
$= \ 48.5 + 1 \times -1.225 = 47.275$
$t_2 = 6 + 1 = 7$
$T_3 = T_2 + hf(t_2, T_2) = 47.275 + 1 \times f(7, 47.275)$
$= 47.275 + 1 \times -0.963750 = 46.31125$
$t_3 = 7 + 1 = 8$
$\therefore T = 46.3$°C (3 SF)

11)

$t_0 = 2, B_0 = 970$ using $h = 1$
$B_1 = B_0 + hf(t_0, B_0) = 970 + 1 \times f(2, 970)$
$= 970 + 1 \times 12.401511529 \ldots \ = 982.401511529 \ldots$
$t_1 = 2 + 1 = 3$
$B_2 = y_1 + hf(t_1, B_1) = 982.401511529 \ldots + 1 \times f(3, 982.401511529 \ldots)$
$= 12.401511529 \ldots + 1 \times 10.177701124 \ldots = 992.579212653 \ldots$
$t_2 = 3 + 1 = 4$
$\therefore B =$£992.58

12) a)

$x_0 = 0$, $y_0 = 1$ so want $x = 0.3$ in three steps: $h = 0.1$

$y_1 = y_0 + h\mathrm{f}(x_0) = 1 + 0.1 \times \mathrm{f}(0)$

$= 1 + 0.1 \times 0 = 1$

$x_1 = 0 + 0.1 = 0.1$

$y_2 = y_1 + h\mathrm{f}(x_1) = 1 + 0.1 \times \mathrm{f}(0.1)$

$= 1 + 0.1 \times 0.0998334167 \ldots = 1.009983342 \ldots$

$x_2 = 0.1 + 0.1 = 0.2$

$y_3 = y_2 + h\mathrm{f}(x_2) = 1.009983342 \ldots + 0.1 \times \mathrm{f}(0.2)$

$= 1.009983342 \ldots + 0.1 \times 0.1986693308 \ldots = 1.029850275 \ldots$

$x_3 = 0.2 + 0.1 = 0.3$

$\therefore y = 1.030$ (3 DP)

12) b)

Since $\frac{\mathrm{d}y}{\mathrm{d}x} = \sin x$ then $y = -\cos x + c$

So y is concave, so the tangents lie below the curve, so each step gives a value that is slightly too small, so it is an under-estimate.

Alternatively:

Integrating: $y = -\cos x + c$

At $(0, 1)$ then $1 = -\cos 0 + c \therefore c = 2$

$y = -\cos x + 2$

$y|_{x=0.3} = -\cos 0.3 + 2 = 1.044664 \ldots > 1.30$

$\therefore$ under-estimate

13)

$x_0 = 3$, $y_0 = a$

$y_1 = y_0 + h\mathrm{f}(x_0, y_0) = a + 0.2 \times \mathrm{f}(3, a)$

$\therefore a + 0.2 \times \frac{6-a}{3a} = 4.450$

$\Rightarrow 3a^2 + 1.2 - 0.2a = 13.35a$

$\Rightarrow 3a^2 - 13.55a + 1.2 = 0$

$\therefore a = 0.09036897 \ldots$ or $a = 4.426298678 \ldots$

But $a > 2$ and so $a = 4.426$

14) a)

$x_0 = 1, y_0 = 0.8$

$y_1 = y_0 + \frac{1}{k}\mathrm{f}(x_0, y_0) = 0.8 + \frac{1}{k} \times \mathrm{f}(1, 0.8)$

$= 0.8 + \frac{1}{k} \times \sqrt{1.6 + 1}$

$\therefore 0.8 + \frac{1}{k} \times \sqrt{2.64} = 1.1250$

$\frac{1}{k} = \frac{1.1250 - 0.8}{\sqrt{2.64}} = 0.2000236728 \ldots$

$\therefore$ Since k is an integer $\frac{1}{k} = \frac{1}{5}$

14) b)

$y_1 = 0.8 + \frac{1}{k} \times \mathrm{f}(1, 0.8) = 0.8 + 0.2 \times \sqrt{2.64} = 1.124961536 \ldots$

$x_1 = 1 + 0.2 = 1.2$

$y_2 = y_1 + \frac{1}{k}\mathrm{f}(x_1, y_1) = 1.124961536 \ldots + 0.2 \times \mathrm{f}(1.2, 1.124961536 \ldots)$

$= 1.124961536 \ldots + 0.2 \times 1.914559599 \ldots = 1.507873456 \ldots$

$x_2 = 1.2 + 0.2 = 1.4$

$\therefore (x_2, y_2) = (1.4, 1.508)$

15) a)

At each point (x_n, y_n), the derivative $\frac{\mathrm{d}y}{\mathrm{d}x} = x_n + y_n$ gives the gradient of the tangent line. The formula moves h units horizontally and then $h \times$ gradient vertically to obtain the next point (x_{n+1}, y_{n+1}) which is the end of a straight-line segment tangent to the curve at the previous point.

15) b)

$x_0 = 0, y_0 = 1$

$y_1 = y_0 + h\mathrm{f}(x_0, y_0) = 1 + 0.5 \times \mathrm{f}(0, 1)$

$= 1 + 0.5 \times 1 \ = 1.5$

$x_1 = 0 + 0.5 = 0.5$

$y_2 = y_1 + h\mathrm{f}(x_1, y_1) = 1.5 + 0.5 \times \mathrm{f}(0.5, 1.5)$

$= 0.5 + 0.5 \times 2 = 2.5$

$x_2 = 0.5 + 0.5 = 1$

$\therefore y = 2.5$

15) c)

$\frac{dy}{dx} - y = x$ has integrating factor $e^{\int -1 dx} = e^{-x}$

$e^{-x}\frac{dy}{dx} - e^{-x}y = xe^{-x}$

$ye^{-x} = \int xe^{-x}\,dx$

Use Integration by Parts:

$u = x$ $\qquad$ $\frac{dv}{dx} = e^{-x}$

$\frac{du}{dx} = 1$ $\qquad$ $v = -e^{-x}$

$\therefore \int xe^{-x}\,dx = -xe^{-x} + \int e^{-x}\,dx = -xe^{-x} - e^{-x}(+c)$

So that:

$ye^{-x} = -xe^{-x} - e^{-x} + c$

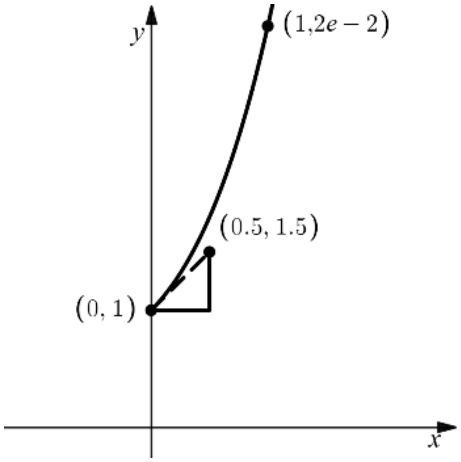
$y = -x - 1 + ce^{x}$

When $y(0) = 1$ then $1 = 0 - 1 + c \therefore c = 2$

$y = 2e^{x} - x - 1$

$\therefore\ y(1) = 2e - 2$

15) d)



16) a)

$\frac{dy}{dt} = kt$

But since $24 = 4k \therefore k = 6$

$\therefore \frac{dy}{dt} = 6t$

16) b)

$t_0 = 1,\ y_0 = 1.5$

$y_1 = y_0 + hf(t_0) = 1.5 + 0.25 \times f(1)$

$= 1.5 + 0.25 \times 6 \ = 3$

$t_1 = 1 + 0.25 = 1.25$

$y_2 = y_1 + hf(t_1, y_1) = 3 + 0.25 \times f(1.25)$

$= 3 + 0.25 \times 7.5 = 4.875$

$t_2 = 1.25 + 0.25 = 1.5$

$\therefore y = 4.88$ GB (2 DP)

17)

$\frac{dN}{dt} = 0.08N - 500e^{-t}$

$t_0 = 1,\ N_0 = 4700$

$N_1 = N_0 + hf(t_0, N_0) = 4700 + 0.5 \times f(1, 4700)$

$= 4700 + 0.5 \times 192.060279414 \ldots \ = 4796.030139707 \ldots$

$t_1 = 1 + 0.5 = 1.5$

$N_2 = N_1 + hf(t_1, N_1) = 4796.030139707 \ldots + 0.5 \times f(1.5, 4796.030139707 \ldots)$

$= 4796.030139707 \ldots + 0.5 \times 272.117331102 \ldots = .088805258 \ldots$

$t_2 = 1.5 \ + 0.5 = 2$

$\therefore N = 4932$ views

18)

$\frac{dW}{dt} = 0.5 - 2\sin^2\left(\frac{\pi}{12}t\right) - 0.01W$

$t_0 = 0,\ W_0 = 100$

$W_1 = W_0 + hf(t_0, W_0) = 1000 + 3 \times f(0, 1000)$

$= 1000 + 3 \times -0.5 \ = 98.5$

$t_1 = 0 + 3 = 3$

$W_2 = W_1 + hf(t_1, W_1) = 98.5 \ + 3 \times f(3, 98.5)$

$= 98.5 \ + 3 \times -1.485 = 94.045$

$t_2 = 3 \ + 3 = 6$

$W_3 = W_2 + hf(t_2, W_2) = 94.045 + 3 \times f(6, 94.045)$

$= 94.045 + 3 \times -2.44045 = 86.72365$

$t_3 = 3 + 6 = 9$

$\therefore W = 86.7$ litres